

Mind the Mistake

Half of algebra marks are lost to a handful of repeated slips. Learn to spot them, explain what went wrong, and write the correct expression.

The classic errors

1. **Forgetting the power.** $-3p \times -5p = +15p^2$, not $-15p$. Multiplying p by p gives p^2 .
2. **Dropping the middle term.** $(5m + 6n)^2$ is $25m^2 + \mathbf{60mn} + 36n^2$, not $25m^2 + 36n^2$.
3. **Sign slips.** $-3p(-5p + 2q) = 15p^2 - 6pq$. Each term inside keeps its sign rule.
4. **Combining unlike terms.** $5w^2$ and $6w$ cannot be added; $2a^2b$ and $3ab^2$ are different terms.
5. **Order of operations.** $y + 2(y + 2) = y + 2y + 4 = 3y + 4$ — multiply the bracket first, then add y .

Best habit: expand fully first, then collect like terms at the end. Most mistakes happen when steps are skipped or combined too early.

Prerequisite refresher: "Like terms" have exactly the same letters and powers. Only like terms can be added or subtracted into one term.

Where marks slip away

Trap 1 — $(a + b)^2 \neq a^2 + b^2$. The missing $2ab$ is the single most common error.

Trap 2 — Negative $\times$ negative. $(-3p)(-5p)$ is positive: $+15p^2$.

Trap 3 — Adding unlike terms. $5w^2 + 6w$ stays as it is; you cannot make $11w^2$.

Trap 4 — Multiplying only some terms. In $y + 2(y + 2)$, the 2 multiplies both y and 2 inside the bracket, but not the y outside.

Slip 5 — Check by substituting a number (e.g. put $a = 2$, $b = 3$) into both your answer and the original to confirm they match.

Model answers — write them like this

Example A. Spot and correct: $-3p(-5p + 2q) = -3p + 5p - 2q$.

Mistake: The $-3p$ was not multiplied — it was just added to the terms.

Correct: $-3p \times (-5p) = 15p^2$; $-3p \times 2q = -6pq$.

$$\therefore 15p^2 - 6pq$$

Example B. Spot and correct: $2(x - 1) + 3(x + 4) = 5x + 3$.

Mistake: The constants were added wrongly: $-2 + 12 = 10$, not 3.

Correct: $2x - 2 + 3x + 12 = 5x + 10$.

$$\therefore 5x + 10$$

Example C. Spot and correct: $(5m + 6n)^2 = 25m^2 + 36n^2$.

Mistake: The middle term $2 \times (5m) \times (6n) = 60mn$ was left out.

Correct: $(5m)^2 + 2(5m)(6n) + (6n)^2 = 25m^2 + 60mn + 36n^2$.

$$\therefore 25m^2 + 60mn + 36n^2$$

Example D. Spot and correct: $2a^3 + 3a^3 + 6a^2b + 6ab^2 = 5a^3 + 12a^2b^2$.

Mistake: $6a^2b$ and $6ab^2$ are not like terms, so they cannot be combined into $12a^2b^2$.

Correct: $2a^3 + 3a^3 = 5a^3$; the other two stay separate.

$$\therefore 5a^3 + 6a^2b + 6ab^2$$

Next: [try the worksheet](#), then [check the answer key](#).